

CAP 4611 Assignment Submission Instructions

1 The Assignments

- The code and data for assignment X is in **aX.zip** on the course website, under **Assignments/Notes**.
- Modifications/updates/clarifications after the assignment is first put online will be marked in **red**. Any major updates will be announced on webcourses.

1.1 Groups

- You can work on your own or in a group of 2, **except for the first assignment** which must be done alone.
- You can have different groups on different assignments.
- If you work in a group, only one of you needs to submit the assignment, but you must indicate who your group members are.

1.2 Grading

- All homework assignments have the same weight.
- We try to provide grades within 10 days of the last possible submission date, i.e., Wednesday after the deadline (see late “class” policy in the syllabus).
- If you think there is a problem with your grade, you have one week to raise a concern from the time that your grades were posted. After that, your grade is final.
- To raise a concern, contact the TA who graded your assignment.

2 Submission Format

- **Submit your assignment as a single PDF file.** Do not submit multiple files.
- Name your file as **aX_student1_student2.pdf** where X is the assignment number, and **student1** and **student2** are the last names of the students in your group (if you are working alone, just use your last name).
- If you are working in a group, only one of you needs to submit the assignment, but you must indicate who your group members are in the document.
- Put your name(s) and student number(s) on the first page, as a backup.

- Your submission needs to be organized sequentially using the (sub-)section numbers in the assignment.
- Figures AND relevant code need to be placed in the appropriate location in the document. **Please only submit the code you wrote**, not the provided “boilerplate” code that we wrote for you. Make sure to include all outputs generated via code in your submission as well (e.g., the generated plots, runtime, error, etc.)
- If you use information from students outside your group or from other sources, cite this at the start of each question. **You will receive a mark of 0 for the assignment (and possibly further consequences) if you are found copying from other sources without citation.**
- Make sure your answers are easy to find, clearly written and easy for the TA to understand. TAs can reduce your mark if your answers aren’t clear or difficult to understand (even if correct).
- To reduce the chance of this happening, you should prepare your report digitally, though this is not required. Read Section 4 for more details on how to write your answers and include code via $\text{\LaTeX}$. If you are not comfortable with $\text{\LaTeX}$, you can use any other document preparation system (e.g., Microsoft Word, Google Docs, etc.) as long as you can export it to PDF. **You will receive +10 bonus points if you use any digital document preparation system to prepare your submission, provided the entire assignment is written digitally** (i.e., no handwritten parts). If you decide to write your answers by hand, you still need to include your code in the PDF file (**no screenshots**). You risk losing marks if your code cannot be copied and pasted easily and needs to be re-typed.

1.2 Matrix Algebra Rules

Assume that $\{x, y, z\}$ are $n \times 1$ column vectors and $\{A, B, C\}$ are $n \times n$ real-valued matrices, $\mathbf{0}$ is the zero matrix of appropriate size, and I is the identity matrix of appropriate size. **State whether each of the below is true in general** (you do not need to show your work).

1. $x^T y = \sum_{i=1}^n x_i y_i$.

Answer: 234567

2. $x^T x = \|x\|^2$.

Answer: 234567

3. $x^T x = x x^T$.

Figure 1: An example of a legible answer format.

3 Submitting Your Assignment

- **Submit your assignments as a single PDF to webcourses.**
- If you work in a group, **only hand in one assignment.**
- Assignments are due at **11:55pm ET** on the submission deadline (same if you are using the late class policy, e.g., 11:55pm on Monday for one late class or 11:55pm on Wednesday for 2 late classes).

- **Important:** Submitting the wrong assignment or corrupted files will result in 50% of the assignment mark being deducted. Not including answers in the correct location in the PDF file can also result in 50% of the assignment mark being deducted.

4 Writing Your Answers with L^AT_EX

Although not necessary to use L^AT_EX, to help you with this, we will put the L^AT_EX file used to generate the assignment PDF on the webpage. The file is, e.g., `a1.tex`. You can use the `\begin{answer}` environment, or the slightly-shorter `\ans{}` command, to write your answers (appearing in green).

4.1 Including code

There are several options for including the code into your answer file. When you submit code, **please only submit the code you wrote**, not the provided “boilerplate” code that we wrote for you.

4.1.1 `minted`

The nicest option is the `minted` package (package documentation, Overleaf’s description). If you’re compiling your document on Overleaf, it should Just Work™ to add `\usepackage{minted}` to your preamble, then include something like

```
\begin{minted}{python}
import numpy as np
def sort_really_really_fast(ary):
    while not np.all(np.diff(ary) >= 0):
        np.random.shuffle(ary)
\end{minted}
```

which will end up looking like

```
import numpy as np
def sort_really_really_fast(ary):
    while not np.all(np.diff(ary) >= 0):
        np.random.shuffle(ary)
```

You can also use `\inputminted[firstline=12,lastline=23]{python}{code/filename.py}` to pull code in directly from your code file – but double-check that you’re including all the relevant bits and not too much boilerplate if you do it this way.

To use `minted` locally, you’ll need to `pip install pygments` and run `latexmk/pdflatex` with the `--shell-escape` flag. Getting the paths right might be annoying if you’re using an IDE instead of the command line, so you might want to consider another option below. If you do this, you should probably comment out the `\usepackage{minted}` in the preamble (or else you’ll still get an error about requiring `-shell-escape`).

4.1.2 `listings`

If you’re having trouble with `minted`, a less-nice but easier-to-get-working option is the `listings` package (Overleaf description). To use this, just put `\usepackage{listings}` in the preamble, then use

```
\begin{lstlisting}[language=Python]
import numpy as np
def sort_really_really_fast(ary):
    while not np.all(np.diff(ary) >= 0):
        np.random.shuffle(ary)
\end{lstlisting}
```

which will end up looking like

```
import numpy as np
def sort_really_really_fast(ary):
    while not np.all(np.diff(ary) >= 0):
        np.random.shuffle(ary)
```

It's fine, but not as pretty as `minted` was. If you want, you can add some colour to it; see Overleaf's example.

You can use `\lstinputlisting[language=Python, firstline=12, lastline=23]{filename.py}` to include a whole file (again, double-check you're including all your code and not too much boilerplate).

4.1.3 Last resort

If for some reason `listings` is giving you issues, it's okay to just include a screenshot with `\includegraphics`. Please try to make one of the above options work, though – if we're not sure about something in your code from just looking at it, we really don't want to have to re-type it to run it.